The documentation process conversion measures necessary to comply with this revision shall be completed by 15 January 1998

INCH POUND

MIL-PRF-19500/168D
15 October 1997
SUPERSEDING
MIL-S-19500/168C
29 January 1968

PERFORMANCE SPECIFICATION SHEET

SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON
TYPES 2N1771A, 2N1772A, 2N1774A, 2N1776A, 2N1777A, JAN, AND JANTX

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the performance requirements for PNPN, silicon power, reverse-blocking triode thyristors. Two levels of product assurance are provided for each device type as specified in MIL-PRF-19500.

1.2 Physical dimensions. See figure 1 (similar to TO-64).

1.3 Maximum ratings.

Limits	I_O 1/ T_C = +105°C (180° condition angle)	I_{TSM} 2/	V_{GM}	T_{op}	T_{STG}	d_V/d_t 3/ (repetitive)
	A	A	V(pk)	°C	°C	v/µs
Min				-65	-65	5
Max	4.7	60	10	+150	+150	

1/ This average forward current is for a maximum case temperature of +105°C, and 180 electrical degrees of conduction.
2/ Surge rating is non-recurrent and applies only with device in the 'on' conducting state. The peak rate of surge current must not exceed 40 amperes/microsecond during the first 10 µs after switching from the "off" (blocking) state to the "on" (conducting) state. This time is measured from the point where the thyristor voltage has decayed to 90 percent of its initial blocking value.
3/ T_C = -65° to +150°C.

1.4 Primary electrical characteristics.

1.4.1 Primary electrical characteristics (common to all types).

Limits	V_{TM}	I_H	V_{GT} 1/	I_{GT} 1/	t_{off} 1/	t_{on}
	V	mA dc	V dc	mA dc	µs	µs
Min			0.2			
Max	1.85	25	2.0	30	30	5

1/ T_C = -65° to +150°C.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5961

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1.4.2 Individual ratings.

Types	V_{RM}	V_{DM}	V_{RM} (nonrep)
	V(pk) 1/	V(pk)	V(pk) 2/
2N1771A	50	50	75
2N1772A	100	100	150
2N1774A	200	200	300
2N1776A	300	300	400
2N1777A	400	400	500

1/ Values apply for zero or negative gate voltage V_{GK}.
2/ Peak reverse voltage, nonrepetitive, t = 5.0 ms maximum.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-PRF-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-129 - Military Marking.
MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications or specification sheets), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Devices furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list before contract award (see 4.2 and 6.3).

3.2 Associated specification. The individual item requirements shall be in accordance with MIL-PRF-19500 and as specified herein.

3.3 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-PRF-19500.

3.4 Interface requirements and physical dimensions. The interface requirements and physical dimensions for the purpose of interchangeability shall be as specified on figure 1 herein

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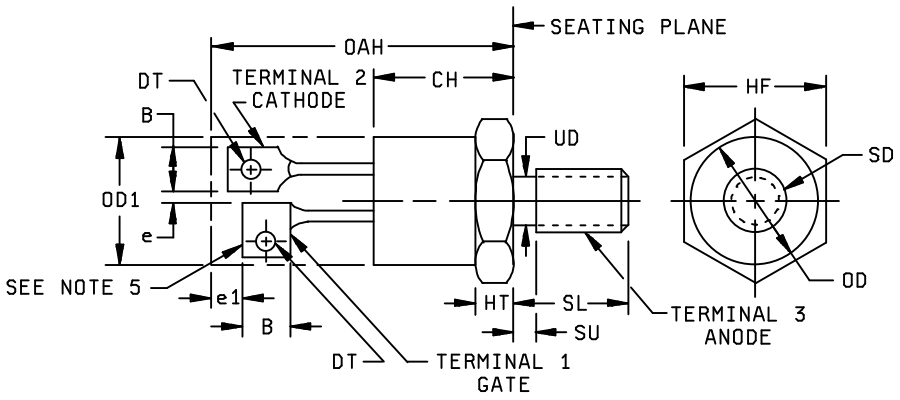

Ltr	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
B	.080	.136	2.03	3.45	2
CH	.300	.400	7.62	10.16	
DT	.040	.075	1.02	1.91	
e	.013	- - -	0.33	- - -	7
e_1	.060	- - -	1.52	- - -	6
HF	.424	.437	10.77	11.10	
HT	.060	.175	1.52	4.45	4
OAH	.700	.855	17.78	21.72	2
OD	.400	- - -	10.16	- - -	4
OD_1	- - -	.424	- - -	10.77	2
SD	.1658	.1697	4.212	4.310	7
SL	.400	.453	10.16	11.51	
SU	- - -	.078	- - -	1.98	
UD	.163	.189	4.14	4.80	

NOTES:
1. Metric equivalents are given for general information only.
2. Device contour except on hex head and noted terminal dimensions is optional within zone defined by OD_1 and OAH, OD_1 not to exceed actual HF.
3. Contour and angular orientation of terminals 1 and 2 with respect to hex portion and to each other are optional.
4. Chamfer or undercut on one or both ends of the hexagonal portion are optional.
5. Square or radius on end of terminal is optional.
6. Minimum difference in terminal lengths to establish datum line for numbering terminals.
7. Pitch diameter - thread 10-32 NF-2A (coated). Reference Handbook H28.

FIGURE 1. Physical dimensions of thyristor types 2N1771A, 2N1772A, 2N1774A, 2N1776A, and 2N1777A.

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3.4.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-PRF-19500, and herein. Where a choice of lead finish is desired, it shall be specified in the acquisition document (see 6.2).

3.4.2 Construction. These devices shall be constructed in a manner and using materials which enable the devices to meet the applicable requirements of MIL-PRF-19500 and this document.

3.5 Marking. Marking shall be in accordance with MIL-PRF-19500.

3.6 Electrical performance characteristics. Unless otherwise specified herein, the electrical performance characteristics are as specified in 1.3, 1.4, and table I herein.

3.7 Electrical test requirements. The electrical test requirements shall be the subgroups specified in 4.4.2 and 4.4.3.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.2).

b. Screening (see 4.3)

c. Conformance inspection (see 4.4).

4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-PRF-19500.

4.3 Screening (JANTX level only). Screening shall be in accordance with MIL-PRF-19500 (Appendix E, table IV), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

Screen (see table IV, appendix E, of MIL-PRF-19500)	Measurement JANTX level
3a	Condition F, 10 cycles
7	Condition C or D for gross leaks
10	Not required
11	I_{RM}, I_{DM}
12d	Condition A, T_C = +145°C ± 3 percent; V_{RM}, V_{DM} = rated (see 1.4.1).
13	Subgroup 2 of table I herein, ΔI_{RM} = 100 percent of initial value or +1.0 mA(pk), whichever is greater. ΔI_{DM} = 100 percent of initial value or +1.0 mA(pk), whichever is greater.

4.4 Conformance inspection. Conformance inspection shall be in accordance with MIL-PRF-19500.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with appendix E, table V of MIL-PRF-19500 and table I herein.

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4.4.2 Group B inspection (JANTX and JANTXV) . Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VIb (JANTX and JANTXV) of MIL-PRF-19500, and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

4.4.2.1 Group B inspection, appendix E, table VIb of MIL-PRF-19500.

Subgroup	Method	Condition
B2	1051	Test condition F, 15 cycles.
B2	1071	Condition C or D for gross leaks.
B2	4066	I_{TSM} = 150 A(pk) (0.5 sine wave); 6 surges per 1 minute; I_O = 4.7 A at rated V_{RRM}; T_C = +65°C; f = 60 Hz; surge duration = 7 ms, minimum.
B3	1040	AC method, bias condition D.
B5		Not applicable.

4.4.3 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in appendix E, table VII of MIL-PRF-19500 and as follows. Electrical measurements (end-points) and delta requirements shall be in accordance with the applicable steps and footnotes of table II herein.

4.4.3.1 Group C inspection, appendix E, table VII of MIL-PRF-19500

Subgroup	Method	Condition
C2	2036	Test condition A, weight 10 pounds, application time = 15 seconds.
C2	1071	Condition C or D for gross leaks.
C3		Not applicable.
C6	1026	T_C = +118°C, ± 5°C, 50 min “on”, I_O = 3 A, T_C uncontrolled, 10 minutes “off”.

.

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows.

4.5.1 Pulse measurements. Conditions for pulse measurement shall be as specified in section 4 of MIL-STD-750.

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TABLE I. Group A inspection.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 1						
Visual and mechanical examination	2071					
Subgroup 2						
Reverse blocking current	4211	AC method, bias condition D; f = 60 Hz; V_{RM} = rated (see 1.4.1)	I_{RM1}		1	mA(pk)
Forward blocking current	4206	AC method, bias condition D; f = 60 Hz, V_{DM} = rated (see 1.4.1)	I_{DM1}		1	mA(pk)
Gate trigger voltage and current	4221	$V_2 = V_D$ = 6 V dc; T_C = + 25°C R_e = 20Ω maximum	V_{GT1} I_{GT1}		2 15	V dc mA dc
2N1771A, 2N1772A, 2N1774A, 2N1776A, 2N1777A		R_L = 140 Ω R_L = 650 Ω R_L = 650 Ω				
Forward “on” voltage	4226	I_{TM} = 14.8 A(pk) (pulse); pulse width = 8.5 ms; maximum; duty cycle = 2 percent maximum	V_{TM}		1.85	V(pk)
Holding current	4201	Bias condition D; V_{AA} = 24 V dc maximum; I_{F1} = 1 A dc; I_{F2} = 100 mA dc; trigger voltage source = 6 V dc; trigger PW = 25 µs (minimum); R_2 = 50 Ω	I_H		25	mA dc
Subgroup 3						
High temperature operation:		T_C = +145°C			2	mA(pk)
Reverse blocking current	4211	AC method, bias condition D; f = 60 Hz; V_{RM} = rated (see 1.4.1).	I_{RM2}		2	mA(pk)
Forward blocking current	4206	AC method, bias condition D; f = 60 Hz; V_{DRM} = rated (see 1.4.1).	I_{DRM2}	0.2		V dc
Gate trigger voltage and current	4221	$V_2 = V_D$ = 6 V dc; T_C = + 125°C R_e = 20Ω maximum, V_2 = rated V_{DM} R_L = 400 Ω	V_{GT2}			

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 3 - Continued						
Low temperature operation:		T_C = -65°C				
Reverse blocking current	4211	AC method, bias condition D; f = 60 Hz, V_{RM} = rated (see 1.4.1)	I_{RM3}		1	mA(pk)
Forward blocking current	4206	AC method, bias condition D; f = 60 Hz; V_{RM} = rated (see 1.4.1)	I_{DM3}		1	mA(pk)
Gate trigger voltage and current	4221	$V_2 = V_D$ = 6 V dc; R_e = 20Ω maximum	V_{GT1} I_{GT2}		2 30	V dc mA dc
2N1771A, 2N1772A		R_L = 140 Ω				
2N1774A, 2N1776A, 2N1777A		R_L = 650 Ω				
		R_L = 650 Ω				
Subgroups 4 and 5						
Not applicable						
Subgroup 6						
Exponential rate of voltage rise	4231	Bias condition D; T_C = +150°C repetition rate = 60 pps test duration = 15 s; C = 10 µF; 50 ≤ R_L ≤ 400 Ω	d_V/d_t		5	µA dc
2N1771A		V_A = 50 V dc	V_D	47		V dc
2N1772A		V_A = 100 V dc		95		V dc
2N1774A		V_A = 200 V dc		190		V dc
2N1776A		V_A = 300 V dc		285		V dc
2N1777A		V_A = 400 V dc		380		V dc

See footnote at end of table.

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TABLE I. Group A inspection - Continued.

Inspection 1/	MIL-STD-750		Symbol	Limits		Unit
	Method	Conditions		Min	Max	
Subgroup 7						
Circuit commutated turn-off time	4224	T_C = +145°C; I_{TM} = 5 A pk; t_{on} = 100 µs; d_i/d_t = 5 A/µs minimum; reverse voltage at t_1 = 2 V minimum; repetition rate = 1 Hz; d_v/d_t = 5 V/µs; gate bias conditions: gate source voltage = 0 V; gate source resistance = 100 Ω	t_{off}			
2N1771A		$V_{DM} = V_{DRM}$ = 50 V(pk); V_{RM} = 50 V maximum			30	µs
2N1772A		$V_{DM} = V_{DRM}$ = 100 V(pk); V_{RM} = 100 V maximum			30	µs
2N1774A		$V_{DM} = V_{DRM}$ = 200 V(pk); V_{RM} = 200 V maximum			30	µs
2N1776A		$V_{DM} = V_{DRM}$ = 300 V(pk); V_{RM} = 300 V maximum			30	µs
2N1777A		$V_{DM} = V_{DRM}$ = 400 V(pk); V_{RM} = 400 V maximum			30	µs
Gate controlled turn-on time	4223	I_{TM} = 5 A dc; V_{GG} = 10 V dc; R_e = 50 Ω; t_{P1} = 15 ±5 µs; 2 A/µs ≤ d_i/d_t ≤ 200 A/µs.	t_{on}		5	µs
2N1771A		$V_A = V_{DRM}$ = 50 V(pk)				
2N1772A		$V_A = V_{DRM}$ = 100 V(pk)				
2N1774A		$V_A = V_{DRM}$ = 200 V(pk)				
2N1776A		$V_A = V_{DRM}$ = 300 V(pk)				
2N1777A		$V_A = V_{DRM}$ = 400 V(pk)				

1/ For sampling plan, see MIL-PRF-19500.

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TABLE II. Groups A, B, and C electrical end-point inspection measurements. 1/ 2/

Step	Inspection	MIL-STD-750		Symbol	Limits		Unit
		Method	Conditions		Min	Max	
1.	Reverse blocking cutoff current	4211	AC method, bias condition D; f = 60 Hz; V_{RM} = rated (see 1.4.1).	I_{RM1}		1	mA(pk)
2.	Forward blocking current	4206	AC method, bias condition D; f = 60 Hz; V_{DM} = rated (see 1.4.1).	I_{DM1}		1	mA(pk)
3.	Reverse blocking cutoff current	4211	AC method, bias condition D; T_C = +125°C; f = 60 Hz; V_{RM} = rated (see 1.4.1)	I_{RM2}		2	mA(pk)
4.	Forward blocking current	4206	AC method; bias condition D; T_C = +125°C; f = 60 Hz; V_{DR} = rated (see 1.4.1)	I_{DM2}		2	mA(pk)
5.	Gate trigger voltage and current	4221	$V_2 = V_D$ = 6 V dc; R_L = 50Ω; R_e = 20Ω maximum	V_{GT1} I_{GT1}		2 15	V dc mA dc
6.	Forward “on” voltage transfer ratio	4226	I_{TM} = 14.8 A(pk) (pulse); pulse width = 8.5 ms maximum; duty cycle = 2 percent maximum	V_{TM}		1.85	V(pk)

1/ The electrical measurements for appendix E, table VIb (JAN and JANTX) of MIL-PRF-19500 are as follows:

Subgroups 2, 3, and 6, see table II herein, steps 1, 2, 3, 4, 5, and 6.

2/ The electrical measurements for appendix E, table VII of MIL-PRF-19500 are as follows:

Subgroups 2, 3, and 6, see table II herein, steps 1, 2, 3, 4, 5, and 6.

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5. PACKAGING

5.1 Packaging. Packaging shall prevent mechanical damage of the devices during shipping and handling and shall not be detrimental to the device. When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Points' packaging activity within the Military Department or Defense Agency, or within the Military Departments' System Command. Packaging data retrieval is available from the managing Military Departments' or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

5.2 Marking. Unless otherwise specified (see 6.2), marking shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-PRF-19500 are applicable to this specification.

6.2 Acquisition requirements. See MIL- PRF-19500.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award of contract, qualified for inclusion in Qualified Products List QPL No.19500 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Columbus, ATTN: DSCC-VQE, 3990 East Broad Street, Columbus, OH 43216-5000.

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - AV, MI
Air Force - 19, 85, 99

Preparing activity:
DLA - CC

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INSTRUCTIONS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-PRF-19500/168D	2. DOCUMENT DATE (YYMMDD) 971015

3. DOCUMENT TITLE
SEMICONDUCTOR DEVICE, THYRISTORS (CONTROLLED RECTIFIERS), SILICON, TYPES 2N1771A, 2N1772A, 2N1774A, 2N1776A, 2N1777A, JAN, AND JANTX

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Alan Barone	b. TELEPHONE (Include Area Code) (1) Commercial (614)692-0510 (2) AUTOVON 850-0510
c. ADDRESS (Include Zip Code) from Defense Supply Center Columbus, ATTN: DSCC-VAT, 3990 East Broad Street, Columbus, OH 43216-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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